# YASSER EL-SAYED M. EL-ALFY

*Researcher | KFUPM*

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Yasser El-Alfy is a researcher at KFUPM working on learning-based perception and 3D SLAM. His work focuses on integrating deep learning with traditional robotics to build robust navigation systems for autonomous vehicles and drones.

**CITATIONS: 21 | H-INDEX: 2 | i10-INDEX: 1**

## EDUCATION

**Doctor of Philosophy in Computer Engineering**
**King Fahd University of Petroleum and Minerals, Saudi Arabia**
2023-present

**Master of Science in Computer Engineering**
**King Fahd University of Petroleum and Minerals, Saudi Arabia**
2021-2022

*CGPA: 3.688*

**Bachelor of Science in Computer & Communication Engineering (BSCCE)**
**Ahlia University, College of Engineering, Bahrain**
2018-2020

*CGPA: 3.99*

**High School**
**Ahlia Schools, King Fahd University of Petroleum and Minerals, Saudi Arabia**
2012-2014

*Cumulative Grade: 99.92%*

## WORKSHOPS

**Advanced Quantum Computing Workshop**
**Saudi Data and Artificial Intelligence Authority (SDAIA)**
11-15 Jan 2026

## PROFESSIONAL EXPERIENCE

**Teaching and Lab Assistant**
**King Fahd University of Petroleum and Minerals (KFUPM)**
2023-present

**Research Intern**
**Center for Communication and Information Technology Research (CCITR), Research Institute, KFUPM**
2020

**Teaching and Lab Assistant** — **Ahlia University** 2018-2020

## RESEARCH STRATEGY

My research focuses on bridging the gap between raw sensor data and high-level spatial understanding. I develop monocular depth estimation models and integrate them into SLAM pipelines to enable 3D mapping without expensive LiDAR hardware.

**Key Focus Areas:**

- Smart Cities
- Autonomous Navigation
- Drone Applications
- Monocular Depth Estimation
- Visual SLAM & Navigation
- Anomaly Detection
- Road Anomaly Characterization
- Edge AI for Robotics
- 3D Mapping
- Scene Understanding

## TECHNICAL SKILLS

**Languages:**

Java, JavaScript, Visual Basic, Python, TensorFlow, Pytorch, HTML, Assembly, Arduino, VHDL, Verilog

**Software:**

Matlab & Simulink, GNU Octave, Sumo Simulator, Cooja/Contiki Simulator, Xilinx, Modelsim, Cisco Packet Tracer, Arduino IDE, mikroc pro, ISIS 7 Professional, Eclipse, Visual Studio, Visual Studio Code, Google Colab, Jupyter Notebook, Microsoft office programs (Word, Excel, PowerPoint, Access), Latex professional editing, Overleaf, Adobe Photoshop, GIMP, Adobe Indesign, DaVinci Resolve, PowerDirector, Kdenlive, Windows Movie Maker, Microsoft Teams, Zoom, Google Meet, Cisco Webex, Blackboard, Moodle

**OS:**

Background in different Operating Systems: Windows, Mac, Linux

**Other Skills:**

Teamwork and communications skills, Research skills, Video and graphical editing

## TEACHING ACADEMIC PROFILE

| | |
|---|---|
| **COE 412: Senior Design Project II** | 2024-2026 |
| **COE 301: Computer Organization** | 2024-2025 |
| **COE 292: Introduction to Artificial Intelligence** | 2022-2024 |
| **ITCS 122: Introduction to Programming Techniques** | 2018-2020 |

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|---|---|
| **ITCS 101: Introduction to Computers & IT** | 2018-2020 |
| **ITCS 224: Data Structures** | 2019-2020 |
| **ITCS 201: Object Oriented Programming I** | 2019-2020 |
| **ITCS 121: Computer Programming** | 2018-2019 |
| **ITCS 222: Visual Programming** | 2018-2019 |
| **ITMS 325: Web Applications Design** | 2018-2019 |

## TALKS & EVENTS

**Multi-Modal Curriculum Federated Learning for Breast Cancer detection** Seminar | 26th November 2025, 4:00PM

*Bldg. 22, Room 119, King Fahd University of Petroleum and Minerals (KFUPM)*

## CERTIFICATIONS

| | |
|---|---|
| **IELTS Certificate: 6.5** | 2020 |
| **ICDL Certificates in Microsoft Office (Word, Spreadsheets, PowerPoint)** | 2018 |
| **Certificates of appreciation for being lab assistant, Ahlia Univ.** | 2018 |
| **Qiyas - General Aptitude Test: 100%** | 2014 |
| **Qiyas - Achievement Aptitude Test: 97%** | 2014 |

## SELECTED PUBLICATIONS

**Enhancing pothole detection and characterization: integrated segmentation and depth estimation in road anomaly systems**
*Systems and Soft Computing (2025), (2025)*

**Monocular based 3D depth estimation and SLAM integration**
*Drone Systems and Applications (2025), (2025)*

**Intelligent Road Anomaly Detection with Real-time Notification System for Enhanced Road Safety**
*Sustainability (2025), (2025)*

**Real-Time On-Device Vision-Based Road Anomaly Detection with Frame Span Interval (FSI) Event Counting**
*IEEE AICCSA (2025), (2025)*

**Dynamic Real-time Learning of Electromagnetic Interference Parameters for Drone Crashing Attack**
*IEEE PIMRC (2025), (2025)*

**A Comparative Study of Transfer Learning and Custom Models with SSD for Face Mask Detection**
*IEEE (2025), (2025)*

**Pothole Detection Under Various Weather Conditions Using YOLOv9**
*Remote Sensing (2025), (2025)*

**Enhancing monocular depth estimation with an advanced encoder-decoder architecture**
*Neural Computing and Applications 37 (29), 24265-24280, (2025)*

**A federated learning approach to banking loan decisions**
*ISNCC (2023), (2023)*

**Distributed Intrusion Detection Systems Based on Deep Learning Techniques and Boosting Ensemble**
*ACM (2023), (2023)*

**Enhancing Monocular Depth Estimation Using an Enhanced Encoder--Decoder Model**
*Authorea Preprints, (2023)*

**Arduino based obstacle awareness vest for blind people**
*ICT-ASET (2022), (2022)*

**MONOCULAR DEPTH ESTIMATION USING DEEP LEARNING FOR 3D SLAM MAPPING IN AUTONOMOUS NAVIGATION**
*KFUPM Master Thesis, (2022)*

**Ultrasonic Projection for the Blind & Visually Impaired: A Proposed Model (Blind-Bat)**
*Ahlia University BSCCE Graduation Project, (2020)*

## PATENTS

**SYSTEM AND METHOD FOR DETECTING AND COUNTING ROAD ANOMALIES** US12445815

## DATASETS

**Road Anomaly Dataset with Depth Ground Truth**
*The first pothole dataset with depth ground truth values, manually annotated for high accuracy.*

**Pothole Detection Under Various Weather Conditions Dataset**
*A dataset with pothole images and generated/simulated various weather conditions.*